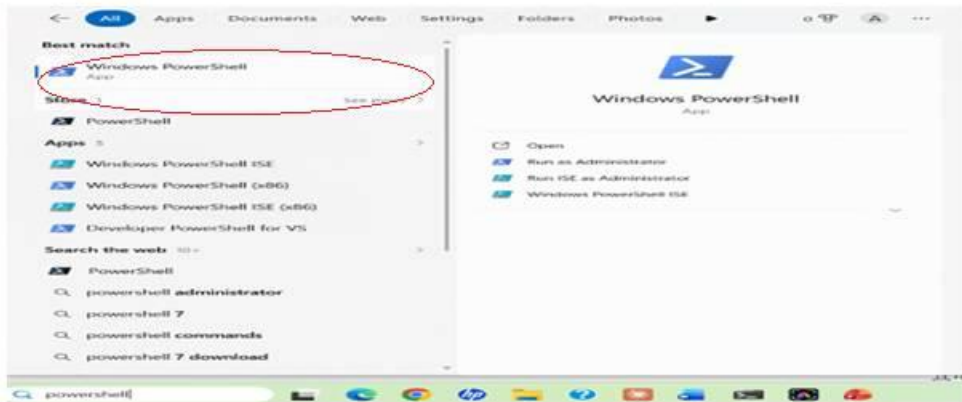


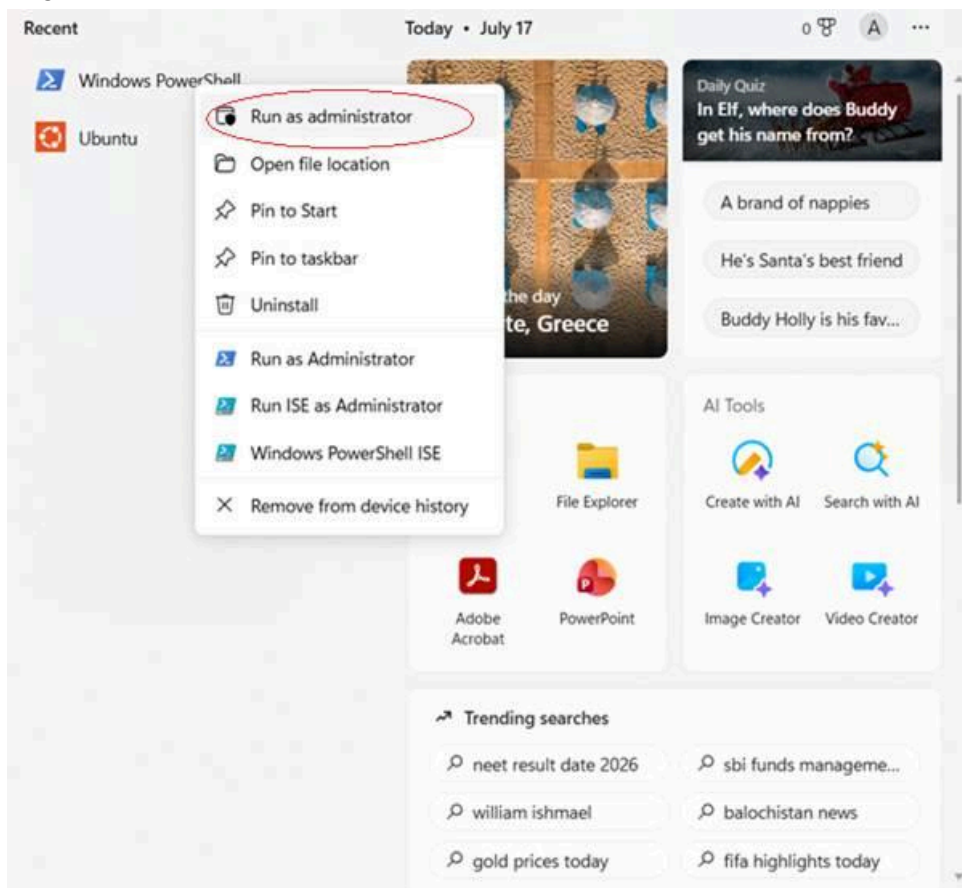
1a.

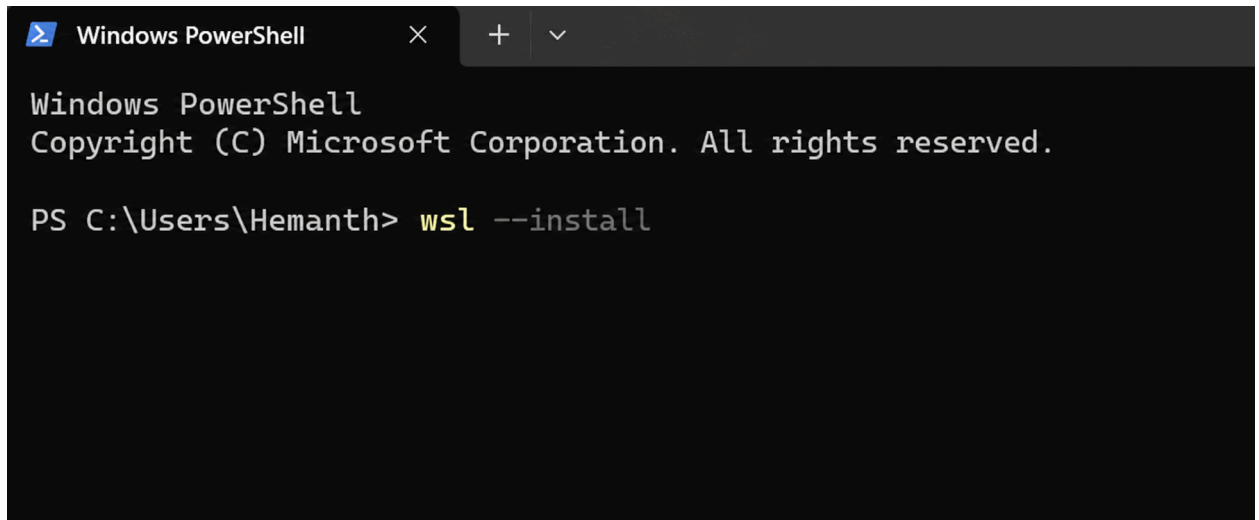
Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



A screenshot of a Windows PowerShell terminal window. The title bar at the top says "Windows PowerShell" with a blue icon on the left and window control buttons (close, maximize, and a dropdown arrow) on the right. The terminal content shows the Windows PowerShell logo, the copyright notice "Copyright (C) Microsoft Corporation. All rights reserved.", and the command prompt "PS C:\Users\Hemanth> wsl --install".

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Hemanth> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: anusha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu:

```

hemanth_venkatesh@LAPTOP-MTJKUJA1:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [293 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [357 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translations-en [76.5 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.1 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [91.1 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [138 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [53.1 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [153 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [43.9 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [255 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [249 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [47.6 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [45.5 kB]
Fetched 2149 kB in 7s (295 kB/s)
74 packages can be upgraded. Run 'apt list --upgradable' to see them.
hemanth_venkatesh@LAPTOP-MTJKUJA1:~$

```

Step 7: Install GCC (C Compiler) :

```

hemanth_venkatesh@LAPTOP-MTJKUJA1:~$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2                gcc-x86-64-linux-gnu    libisl23
  cpp                  libalgorithm-diff-perl libitm1
  cpp-15               libalgorithm-diff-xs-perl liblsan0
  cpp-15-x86-64-linux-gnu libalgorithm-merge-perl libmpc3
  cpp-x86-64-linux-gnu libasan8                 libquadmath0
  dpkg-dev              libc-dev-bin            libstdc++-15-dev
  fakeroot              libc6-dev               libtsan2
  g++                   libcc1-0                libubsan1
  g++-15                libcrypt-dev             linux-libc-dev
  g++-15-x86-64-linux-gnu libdpkg-perl             lto-disabled-list
  g++-x86-64-linux-gnu libfakeroot              make
  gcc                   libfile-fcntllock-perl manpages-dev
  gcc-15                libgcc-15-dev            rpcsvc-proto
  gcc-15-base           libgomp1
  gcc-15-x86-64-linux-gnu libhwasa0

Suggested packages:
  bzip2-doc             gcc-15-doc              gcc-doc
  cpp-doc               gcc-multilib            gcc-15-multilib

```

Step 8: Verify the Installation:

```
hemanth_venkatesh@LAPTOP-MTJKUJA1:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
hemanth_venkatesh@LAPTOP-MTJKUJA1:~$
```

Step 9: Create Your C Program

Example: nano process.c

Step 10: Compile the Program:

```
hemanth_venkatesh@LAPTOP-MTJKUJA1:~$ nano fork1.c
hemanth_venkatesh@LAPTOP-MTJKUJA1:~$ gcc fork1.c
hemanth_venkatesh@LAPTOP-MTJKUJA1:~$ ./a.out
Parent Process
Parent Process id is 872
My Parent Process id is 401
Child Process
Child Process id is 873
Parent Process id is 400
hemanth_venkatesh@LAPTOP-MTJKUJA1:~$
```